

Drone Flight Control with Reinforcement Learning

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We develop autonomous controllers for a planar drone performing sequential target navigation subject to hard boundary constraints. We compare **REINFORCE**, a continuous policy-gradient method theoretically offering finer control, with **SARSA**, a discrete value-based approach prioritising training stability. We demonstrate that while REINFORCE suffers from high variance and optimisation instability, SARSA successfully converges to a robust, boundary-aware policy.

I. INTRODUCTION

We study the autonomous navigation of an underactuated system, planar drone (x, y, θ) controlled by two upward thrusts (T_1, T_2) . Since lateral motion requires prior rotation, actions exhibit delayed kinematic effects that physics-based controllers often struggle to manage. To address this, we employ Reinforcement Learning (RL) to derive anticipatory policies capable of momentum management and collision avoidance for sequential target acquisition within hard boundaries. [1] We compare two RL approaches. **REINFORCE**, a continuous policy gradient method using a shallow neural network, and **SARSA**, a discrete value-based method using tabular state-action values. We demonstrate that while REINFORCE only achieves temporary hover stability, SARSA converges to a robust boundary-aware policy which outperforms a heuristic controller in target acquisition by 28% and path efficiency by 24%.

II. REINFORCE

We implement REINFORCE [2], a policy gradient method that directly optimises a stochastic policy $\pi_\theta(a|s)$. The parameters θ are updated via gradient ascent on the expected return $J(\theta)$, approximated by sampling trajectories:

$$\theta \leftarrow \theta + \alpha G_t \nabla_\theta \log \pi_\theta(a_t|s_t) \quad (1)$$

where α is the learning rate and G_t is the discounted return, calculated over a full episode of simulation.

We use a neural network for the policy, with one hidden layer (32 units, ReLU) and **Gaussian** outputs (μ, σ) for thrust and roll ϕ , mapped via sigmoid and tanh respectively to enforce kinematic constraints. The policy self-regulates exploration by learning σ , which governs the stochasticity of sampled trajectories.

REINFORCE is inherently high-variance, so we employed three stabilisation techniques. **Input Normalisation** to $(0, 1)$ prevents feature bias. **Batch stabilisation** of discounted returns ($\mu(G) = 0, \sigma(G) = 1$) addressing asymmetric reward distributions (e.g., with all-negative returns, even good actions are penalised). **Xavier Initialisation** to stabilise gradients.

We begin with a hover curriculum, with one permanent target at $(0, 0)$. Rewards are shaped to encourage tar-

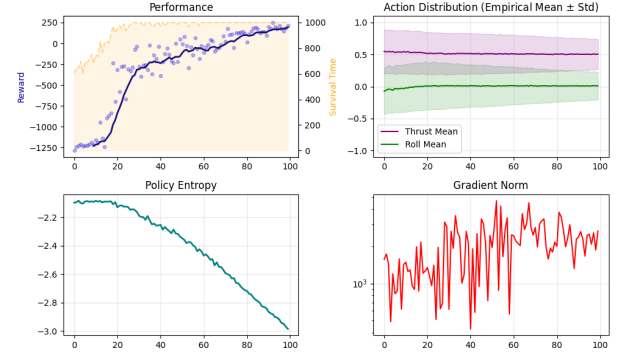


FIG. 1: Training diagnostics. TL: cumulative rewards (blue), survival time (yellow). TR: sampled actions. BL: average sigma (exploration rate). BR: gradient norms.

get proximity and penalise instability. In order to overcome the return smoothing of REINFORCE, we reward intention rather than outcome and localise returns with a low discount factor (0.5). Introducing a positive radius to a distance-to-target reward creates a basin of attraction around the target. We penalise actions which amplify drift, using asymmetry to avoid catastrophic oscillation from overcorrection. $r_{drift} = \min(0, -v_x \cdot \phi)$. We note that **Actor-Critic** (AC) [2] is a more principled approach to the same objective and should be investigated in future work.

Training diagnostic plots (Figure 1) show that monotonic increases in both cumulative rewards and survival time are correlated with gradual exploration decay and decrease in gradient variance. As the network settles into a consistent parameterisation, thrust and roll outputs converge with decreasing variance to counteract gravity. When tested, the agent approaches targets, acquiring them with long enough episode length. The agent learned to stabilise temporarily near targets.

In the subsequent target seeking phase, the agent failed to navigate consistently to more than one target. Both **Weight Decay** to escape the hover extrema, and a **Value Baseline** using a target seeking SARSA model, were unsuccessful as training failed to stabilise. The agent learned stable hover behaviour, however learning to navigate from here would require either significant gradual unlearning, or better variance management.

III. TABULAR SARSA

Tabular methods offer superior stability and interpretability compared to deep reinforcement learning. We employ SARSA [3], an on-policy temporal difference algorithm that updates Q-values as:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha [r_{t+1} + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t)] \quad (2)$$

where α is the learning rate and γ is the discount factor. Unlike off-policy Q-learning, SARSA estimates the value of the current ϵ -greedy policy rather than the *optimal* policy, encouraging conservative behaviour that is well-suited to hard-boundary environments.

a. State Discretisation To enable tabular learning, the continuous state vector comprising target-relative position (d_x, d_y) , velocity (v_x, v_y) , pitch (θ) , and angular rate (ω) is discretised using a non-uniform binning strategy. Target-relative positions (d_x, d_y) utilise a finer granularity of 7 bins with resolution increasing near the target. We define asymmetric inner thresholds where strict vertical bounds of $|d_y| \leq 0.05$ ensure precise altitude hold while relaxed lateral bounds of $|d_x| \leq 0.105$ (slightly outside target radius $R = 0.10$) facilitate anticipatory damping. The dynamic states $(v_x, v_y, \theta, \omega)$ employ coarser ternary bins with tight thresholds to maintain high sensitivity for momentum and attitude control. Additionally, a binary *danger flag* activates when the drone is within 0.1 units of the boundary. This yields a tractable state space of $|\mathcal{S}| = 7,938$.

b. Macro-Action Six heuristic actions map to continuous thrust pairs $(T_1, T_2) \in [0, 1]$: **hover** (altitude hold with pitch stabilisation), **boost up/down** (vertical thrust modulation with level attitude), **tilt left/right** (pitch targeting to $\theta_{opt} = 0.28$ rad with adaptive vertical thrust dependent on d_y), and **arrest motion** (damping of linear and angular velocities).

c. Reward Structure We balance sparse terminal signals target hit reward and crash penalty with dense shaping terms. These include a scaled progress bonus to guide navigation, a near-boundary penalty aligned with the *danger flag* for immediate negative feedback, and a timestep cost to incentivise speed and path efficiency. Weights were iteratively tuned based on reward distribution logs and observed flight behaviours to prevent imbalances.

d. Curriculum Training Training follows three progressive stages. **Stage 0** confines targets to the safe central region to facilitate initial trajectory learning. **Stage 1** expands target generation to the full environment.

From this stage, we enable action repetition ($n_{step} = 3$) to reduce control jitter and facilitate credit assignment. **Stage 2** significantly increases the crash penalty to refine boundary-aware policies. Stage advancement is gated by rolling average target acquisitions. Episodes terminate upon all targets acquired, boundary violation, or time-out.

IV. RESULTS

We evaluate controllers over 100 episodes with 5 random targets (Table I). Performance is assessed by *Target Hits* (average acquisitions), *Crash Rate* (boundary collision%), *Travel Time* (median steps per target), and *Path Efficiency* (ratio of path length to Euclidean distance). In the bounded test, the physics-based heuristic controller achieves the fastest travel time (423 steps) but exhibits a high crash rate (46%). **SARSA Slow** prioritises safety with the lowest crash rate (22%) through a cautious approach at the cost of speed. **SARSA Fast** balances speed and safety, achieving the best path efficiency (1.24) and a 34.00% crash rate. It also demonstrates excellent generalisation to the out-of-sample unconstrained scenario, achieving 7.8 out of 10 target acquisition, 28% more than heuristic (not shown). As expected, the REINFORCE controller achieved only 0.36 target hits, though (not shown) invariably scores highest on stability measures. This suggests that for bounded, unstable dynamics, tabular SARSA can provide robust policy learning despite its discrete nature.

TABLE I: Results (bounded)

	Target Hits	Crash Rate	Travel Time	Path Eff.
Heuristic	4.20	46.00	423.50	1.64
SARSA Slow	4.52	22.00	685.00	1.32
SARSA Fast	4.18	34.00	526.00	1.24
REINFROCE	0.36	58.00	1619.50	1.99

V. CONCLUSIONS

Though both approaches benefit from responsive reward design and curriculum learning, REINFORCE lacks both training stability and fine credit assignment to control the planar drone, requiring additional mechanisms to manage variance. With strategised binning, SARSA is capable of outperforming a physics-based controller, exhibiting superior control through learned path optimisation and safety awareness.

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- [1] J. Mair, Drone flight controller github repo, <https://github.com/JamieMair/DroneControllerMLiS> (2020).
[2] J. P. Garrahan, E. Gillman, and J. F. Mair, Machine learning in science part 1 phys4035 lecture notes, Unpublished

manuscript (2025).

- [3] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. (The MIT Press, 2018).